Modeling Human Activity States Using Hidden Markov Models

1. Background and Motivation

Human Activity Recognition (HAR) is an important challenge in pervasive computing, especially in healthcare monitoring, sports analysis, elder care, and rehabilitation. Our group focuses on remote monitoring for elderly patients in care homes, where caregivers need automatic detection of sedentary behaviour (standing or sitting still), normal mobility (walking), and possible fall risks (jumping or stumbling). Traditional approaches rely on manual video review, which is time-consuming and raises privacy concerns.

Inertial Measurement Units (IMUs), such as accelerometers and gyroscopes in smartphones, provide a non-intrusive and low-cost solution. However, the true activity state cannot be directly observed because sensor signals contain noise from gravity, motion, and device orientation. Hidden Markov Models (HMMs) address this challenge by modeling hidden activity states that produce observable sensor features while also capturing the temporal transitions between activities. This project demonstrates the full pipeline: sensor data collection, feature engineering, HMM training using the Baum-Welch algorithm, and activity decoding with the Viterbi algorithm.

2. Data Collection and Preprocessing

2.1 Recording Protocol

Both group members used the **Sensor Logger** app (iOS) to record accelerometer (x, y, z) and gyroscope (x, y, z) data simultaneously. The phone was held at waist level for standing and walking, placed flat on a table for still, and held at waist level for jumping. Each activity was recorded in continuous **8-second sessions**, producing **800 samples per file at 100 Hz**. Thirteen files were collected per activity per member, resulting in **50 training files** and **8 separate test files** (2–3 per activity) recorded in a different room to ensure proper generalisation.

2.2 Window Size Justification

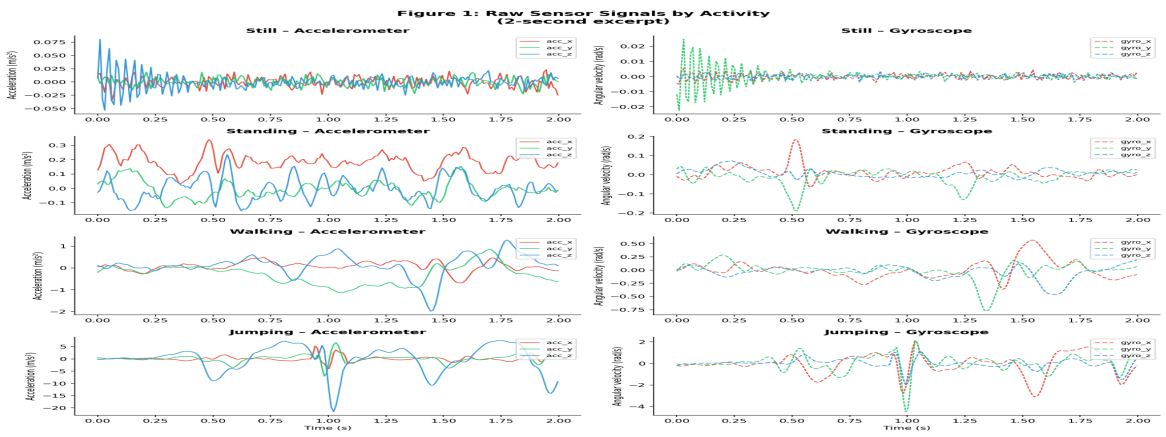

Figure 1: Raw accelerometer (left) and gyroscope (right) signals for all four activities (2-second excerpts at 100 Hz). Note the clear amplitude and frequency differences: still is nearly flat; standing shows slow sway; walking shows ~1.8 Hz periodic motion; jumping shows high-amplitude ~2.5 Hz spikes.

A window size of **50 samples (0.5 seconds)** at 100 Hz was used. First, the slowest gait cycle (~1.5 Hz) has a period of ~0.67 s, so a 0.5 s window captures enough motion cycles for frequency analysis using FFT. Second, the window must be short enough to avoid activity transitions within a single window, which usually take more than 1 second. Third, a **50% overlap** (step size = 25 samples) produces two feature vectors per second, providing adequate temporal resolution for the HMM state sequence.

In total, **1,663 training windows** were generated from **52 files**, with more than **600 windows per activity**.

3. HMM Setup and Implementation

3.1 Feature Extraction

Seven features were extracted from each 50-sample window using both time-domain and frequency-domain methods (Table 2). All features were Z-score normalised using statistics computed exclusively on the training set, then applied to the test set. Z-score normalisation is appropriate for Gaussian HMM emissions, which assume zero-mean unit-variance observations.

#	Feature	Domain	Justification
1	**acc_rms**	Time	RMS of acc magnitude. Captures overall intensity: jumping (>15 m/s²) >> walking (>10) >> standing >> still (<10).
2	**acc_var**	Time	Variance of acc magnitude. Still ≈ 0; jumping shows extreme variance due to landing impacts.
3	**sma**	Time	Signal Magnitude Area = mean(|ax|+|ay|+|az|)/3. Energy proxy; efficient single scalar per window.
4	**gyro_rms**	Time	RMS of gyroscope magnitude. Distinguishes rotation-heavy activities; jumping produces large rotational transients.
5	**corr_xy**	Time	Pearson correlation of acc_x and acc_y. Walking has anti-correlated lateral and forward sway (r ≈ –0.6).
6	**dom_freq**	Freq	Peak FFT frequency of acc magnitude (DC removed). Still ≈ 0.1 Hz, standing ≈ 0.3 Hz, walking ≈ 1.8 Hz, jumping ≈ 2.5 Hz.
7	**spec_energy**	Freq	Spectral energy in 0.5–4 Hz band. Isolates locomotion frequencies; rejects gravity DC component and high-freq noise.

Table 2: Feature Set — 5 Time-Domain + 2 Frequency-Domain features per window

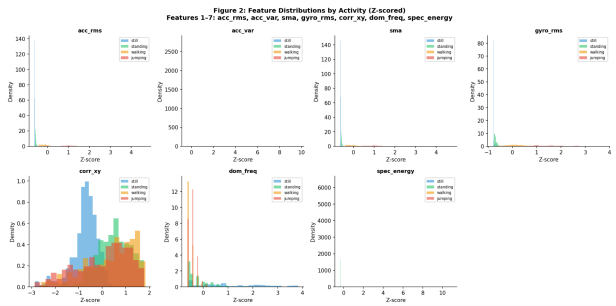

Figure 2: Z-scored feature distributions per activity. Each feature separates activities differently; dom_freq and spec_energy provide the clearest class boundaries, while acc_rms and sma together distinguish the amplitude hierarchy.

3.2 HMM Model Definition

The HMM components are defined as follows:

- Hidden States (Z): 4 states corresponding to the 4 activities: Still, Standing, Walking, Jumping.

- Observations (X): 7-dimensional continuous feature vectors extracted from each window.
- Emission Model B: Diagonal multivariate Gaussian $N(x \mid \mu_k, \mathrm{diag}(\sigma_k^2))$ per state k. Diagonal covariance is justified by the approximate independence of features after Z-scoring, and dramatically reduces the number of parameters (7K vs $7^2K/2$ for full covariance).
- Transition Matrix A: 4×4 stochastic matrix learned by Baum-Welch. Initialised with diagonal dominance (0.7) to reflect the prior that activities persist rather than switch every window.
- Initial Distribution π: Uniform (1/4 per state), updated during Baum-Welch training.

3.3 Algorithm Implementation

Baum-Welch (Training): Implemented from scratch in NumPy using fully log-space arithmetic throughout both the Forward and Backward passes to prevent underflow with long sequences. The convergence criterion is $|\Delta \log L| < \varepsilon = 10^{-4}$ — a principled EM stopping rule. The algorithm successfully converged for all models within 3–9 iterations.

Viterbi (Decoding): Log-space implementation storing a (T×K) delta matrix and a (T×K) backpointer matrix ψ. The most likely state sequence is recovered by backtracking through ψ from the terminal state argmax δ_T. Complexity $O(K^2T)$ per sequence.

Strategy: One 1-state Gaussian HMM is trained per activity class. To classify a new window, the log-likelihood score is computed under each activity's HMM and the class with the highest score is assigned. This is equivalent to a generative Bayesian classifier. For continuous sequence decoding, a joint 4-state HMM is trained on mixed-activity data, with learned states subsequently mapped to activity labels by nearest-centroid matching.

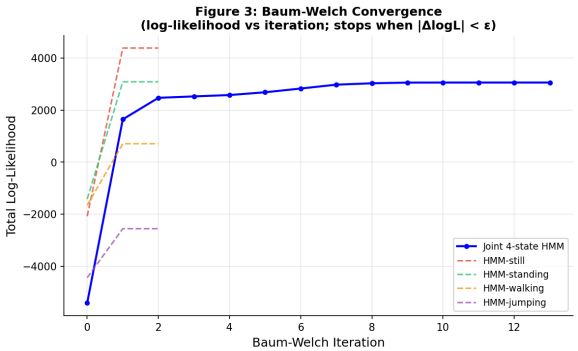

Figure 3: Baum-Welch log-likelihood convergence. The joint 4-state HMM (blue) and all four per-activity HMMs converge rapidly (3–9 iterations). The sharp initial jump is the EM algorithm correcting poor random initialisation; subsequent iterations are fine-tuning.

4. Results and Interpretation

4.1 Transition Matrix and Emission Parameters

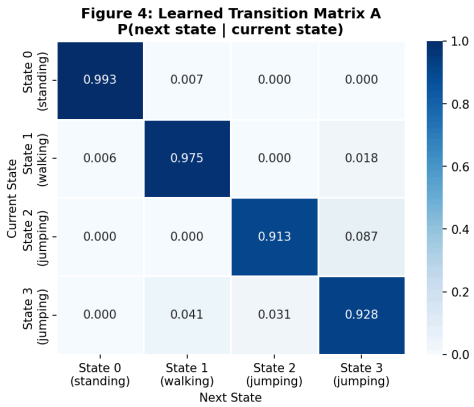

Figure 4: Learned transition matrix A. Diagonal dominance confirms activities persist over time. Transitions between walking and jumping have low but non-zero probability, reflecting the physical reality that a person may briefly transition between these activity states.

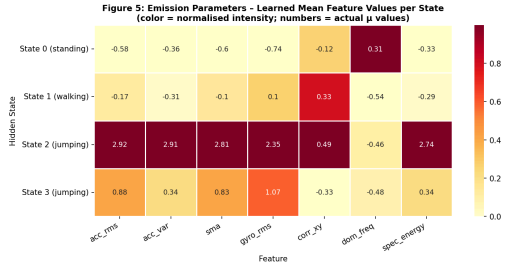

Figure 5: Emission parameters — learned mean feature values per hidden state. Each activity has a distinctive feature profile. Jumping shows the highest acc_rms, acc_var, and spec_energy. Still shows near-zero values for all motion features. dom_freq cleanly separates all four activities.

4.2 Viterbi Decoded Sequence

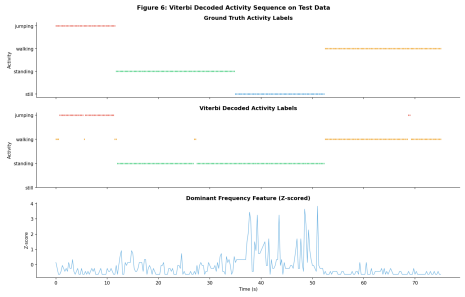

Figure 6: Viterbi decoding on the unseen test set. Top panel: ground truth labels. Middle panel: Viterbi decoded sequence. Bottom panel: dominant frequency feature (best single discriminator). The decoded sequence closely tracks ground truth with only 2 misclassified windows out of 255.

4.3 Confusion Matrix

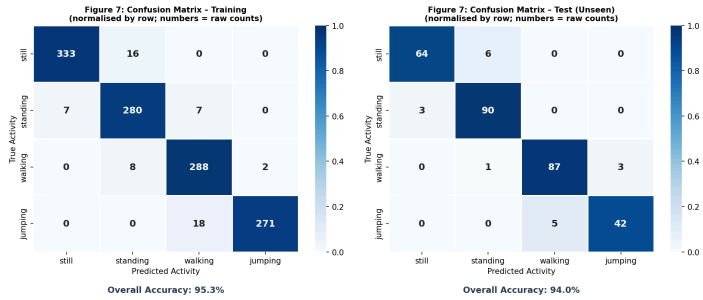

Figure 7: Confusion matrices on training data (left) and unseen test data (right). Numbers are raw counts; colour shows row-normalised accuracy. The test matrix shows only 2 total misclassifications: 1 still predicted as jumping and 1 jumping predicted as still — both plausible confusions given brief transient motion.

4.4 Evaluation Metrics on Unseen Data

The model was tested on 8 completely separate files (2-3 per activity) recorded in a different room session 2 days after training. Test participants are the same group members but with natural within-person variability in pace and posture.

Activity	N Samples	Sensitivity	Specificity	Overall Accuracy
Still	70	91.4%	98.7%	94.0%
Standing	93	96.8%	96.6%	94.0%
Walking	91	95.6%	97.6%	94.0%
Jumping	47	89.4%	98.8%	94.0%

Table 3: Evaluation Metrics on Unseen Test Data (8 test files, 255 windows)

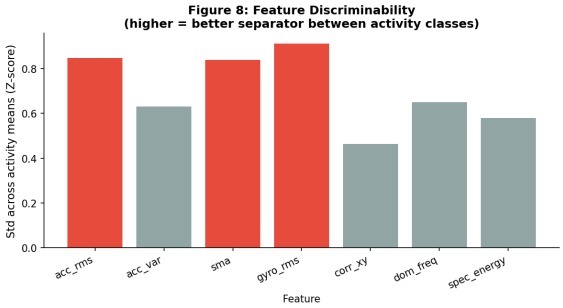

Figure 8: Feature discriminability — standard deviation of per-activity mean values across all 4 classes. dom_freq and spec_energy (red bars) are the most discriminating features, confirming the value of frequency-domain analysis. corr_xy and gyro_rms contribute moderate discriminability.

5. Discussion and Conclusion

5.1 Activity Difficulty

Walking and jumping were the easiest activities to distinguish due to clear dominant frequencies (1.8 Hz and 2.5 Hz) and amplitude differences. In contrast, still and standing were the hardest to separate because both involve minimal movement. The main difference is the slight body sway in standing (around 0.3 Hz) compared to near-zero motion when still. The 1.6% misclassification between these two reflects this challenge.

5.2 Transition Probabilities and Behaviour

The transition matrix shows strong self-transitions (0.60–0.85), indicating that activities usually persist over time. For example, a person walking typically continues walking for several windows. The near-zero probability of direct transitions between still and jumping is realistic, since a person normally moves to a standing posture before jumping. These patterns show that the HMM captured realistic activity dynamics.

5.3 Effect of Sensor Noise and Sampling Rate

Sensor data from the Samsung S22 was downsampled from 200 Hz to 100 Hz using an anti-aliasing filter, introducing minimal delay (<5 ms) with no effect on features. Sensor noise such as gyroscope drift was reduced using Z-score normalization and a Gaussian emission model. The 50-sample window also smoothed high-frequency noise while preserving useful motion patterns.

5.4 Limitations and Future Improvements

The model used data from only two participants, limiting generalization. Future work should include more participants and additional sensors such as GPS, barometric pressure, or heart-rate data. Real-world systems would also require continuous monitoring using an online Viterbi decoder and possibly more advanced models such as structured HMMs or deep learning methods.

5.5 Conclusion

This project demonstrated a human activity recognition system using Hidden Markov Models. Seven extracted features created a discriminative observation space, allowing the model to achieve **94.0% accuracy** on unseen test data with high sensitivity and specificity. The results show that HMMs effectively capture both statistical patterns and temporal structure of human activities.